

# A Complete Course in Physics ( Graphs ) - Extended First Edition

Rajat Kalia

March 09, 2017



# Preface

The First Edition was a collection of tough problems but the file was small , only being 12 pages not meeting the Amazon crieteria of minimum 24 pages for paperback printing. Moreover, the toughness of problems required that answers be provided Intext.

Graphs is a very catchy topic for +1/+2 level and Competitive Exams, it is apparent that books in this particular topic would be written and available soon in the near future.



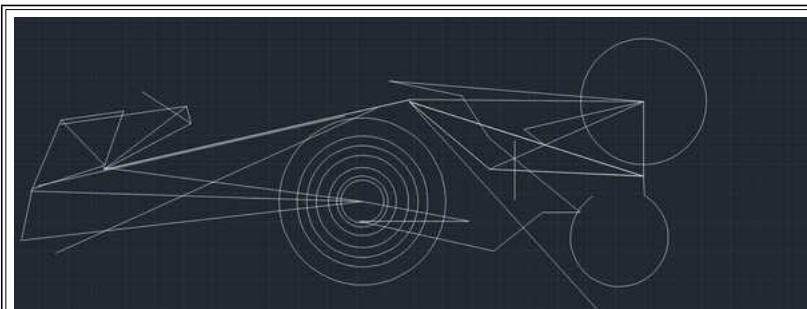
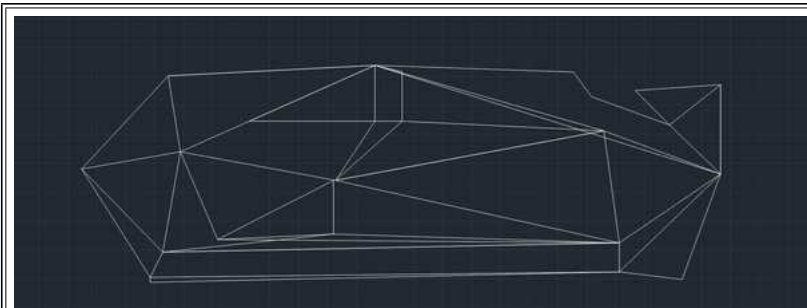
# Acknowledgements

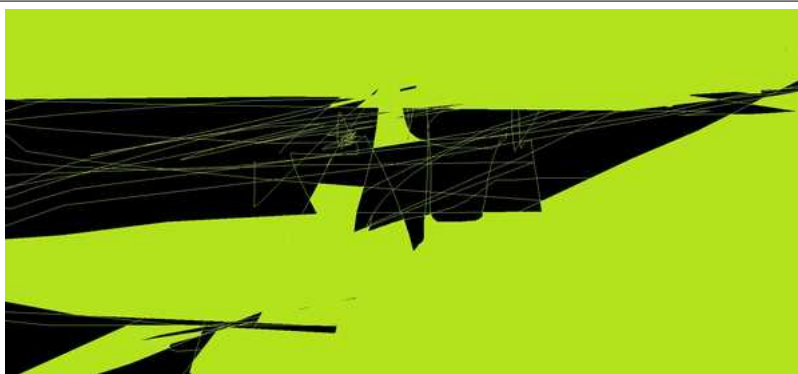
I first of all thank my son Manas Kalia for the love and support.

Today, he is writing an essay on “Myself” for all of us

- 1 My name is Manas.
- 2 I am six years old.
- 3 I study in KGF.
- 4 I like to play ludo.
- 5 I love my family.

I would also like to show few of his designs, he has made while playing on computer.





# Contents

|          |                                                                      |          |
|----------|----------------------------------------------------------------------|----------|
| <b>1</b> | <b>Concepts of Graphs</b>                                            | <b>9</b> |
| 1.1      | The Equations of motion and the origin of Graph Handling . . . . .   | 9        |
| 1.1.1    | The First Equation . . . . .                                         | 9        |
| 1.1.2    | The Second Equation . . . . .                                        | 12       |
| 1.1.3    | The Acceleration-Position Graph Variate . . . . .                    | 15       |
| 1.2      | Other Types of Graphs . . . . .                                      | 16       |
| 1.2.1    | Sign of Acceleration from Position-Time Graph . . . . .              | 16       |
| 1.2.2    | The Average-Velocity / Instantaneous Velocity , Equal Case . . . . . | 17       |
| 1.2.3    | The Velocity-Displacement Case . . . . .                             | 18       |
| 1.2.4    | Motion Under Free Fall due to Gravity . . . . .                      | 20       |
| 1.2.5    | Projectile Motion . . . . .                                          | 21       |
| 1.2.6    | Miscellaneous . . . . .                                              | 22       |
| 1.3      | Question Types . . . . .                                             | 25       |
| 1.3.1    | Passage Type . . . . .                                               | 25       |
| 1.3.2    | Matching . . . . .                                                   | 28       |
| 1.4      | Previous Year Problems IIT . . . . .                                 | 30       |



# Chapter 1

## Concepts of Graphs

### 1.1 The Equations of motion and the origin of Graph Handling

#### 1.1.1 The First Equation

The Equation  $v = \frac{dx}{dt}$  in linear motion implies

- i) The **Slope** of **Position-Time Graph** is **Instantaneous Velocity**.
- ii) The **Area** under the **Velocity-Time Graph** is **Change in Position**.  
{ The second one requires the manipulation ,  $dx = vdt$  i.e.  $\int dx = \int vdt$  }

The equations can be further manipulated to obtain the Speed Time Graph , where

speed = rate of change of distance wrt time

Few of the following examples illustrate this concept:

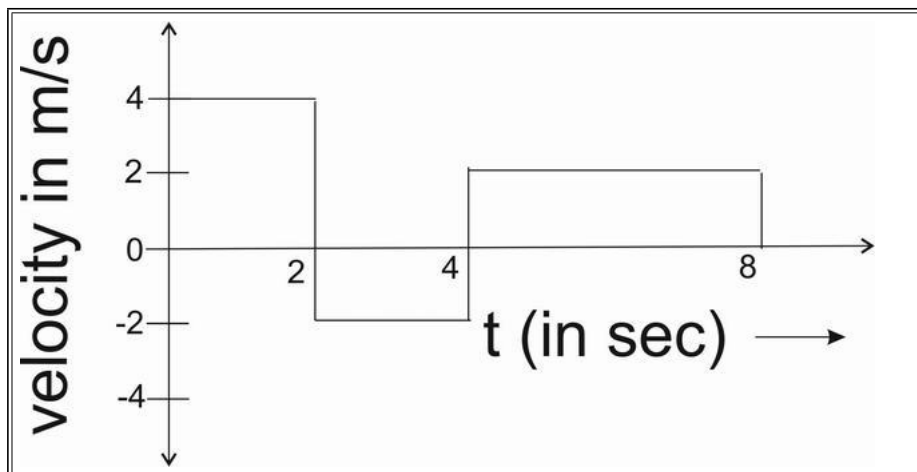
**Example:** On a displacement-time graph, two straight lines make angles of  $30^\circ$  and  $60^\circ$  with the time-axis. The ratio of the velocities represented by them is

- a)  $1 : \sqrt{3}$
- b)  $1:3$
- c)  $\sqrt{3} : 1$
- d)  $3:1$

{ Hint: The velocity in a displacement-time is given by the slope of the curve. Slope =  $\tan(\text{gent})$  of angle of inclination of s-t graph. This gives the respective ratios  $\tan 30^\circ / \tan 60^\circ$

Answer: b) is the correct answer. }

**Example:** A body is moving in a straight line as shown in velocity-time graph. The displacement and distance travelled by body in 8 second are respectively:

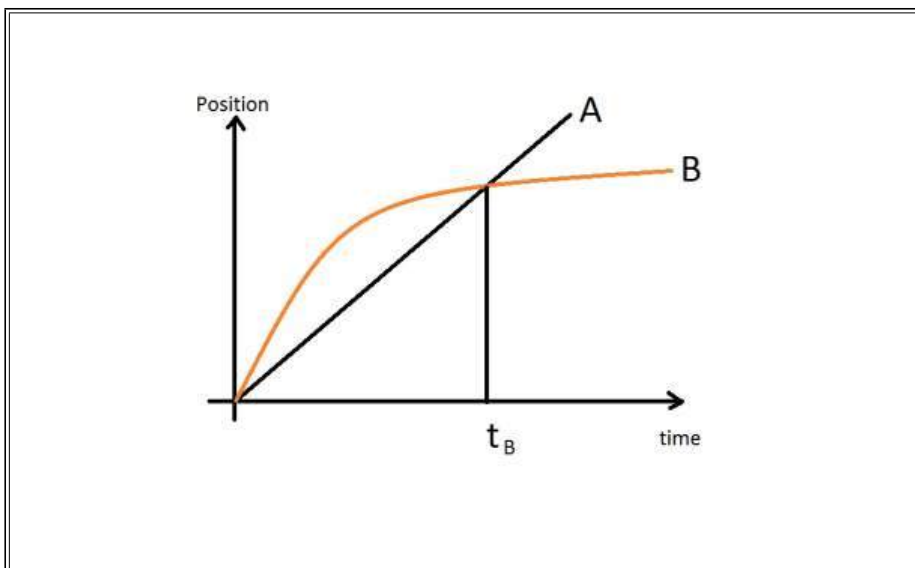


- a) 12 m, 20 m
- b) 20 m, 12 m
- c) 12 m, 12 m
- d) 20 m, 20 m

{ Hint: The displacement in a velocity-time graph is given by the area under the graph with proper signs. From 0s - to 2s , the area is 8m . From 2s - to 4s , the area is -4m . From 4s - to 8s , the area is 8m. Adding these 3 values , we get  $8\text{m} + (-4\text{m}) + 8\text{m} = 12\text{m}$ . } The distance in a v-t graph is given by the absolute area under the graph. So, taking the absolute values of individual area divisions, we get  $8\text{m} + 4\text{m} + 8\text{m} = 20\text{m}$

Answer: a) is the correct answer. }

**Example:** The graph shows position as a function of time for two trains running on parallel tracks. Which statement is true?

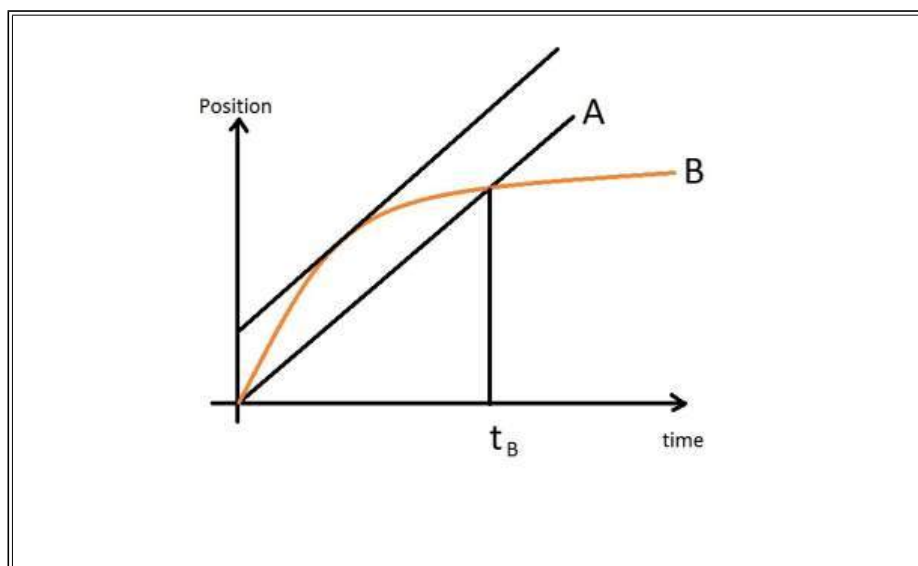


- a) At time  $t_B$  both trains have the same velocity.
- b) Both trains have the same velocity at some time after  $t_B$  .
- c) Both trains have the same velocity at some time before  $t_B$  .
- d) Somewhere on the graph, both trains have the same acceleration.

{ Hint: Depending on the question requirements, we'll have to check all the assertions one by one.

a) In a position time graph, the slope gives velocity. It can be clearly seen that Graph B has a much lower slope than Graph A at time  $t_B$ . So, the assertion is wrong.

b,c) By drawing a line parallel to the line A which is a tangent to Graph B , it can be seen where the two graphs have same slope. It is clear that the graphs have same slope between 0 and  $t_B$  as noted from the figure. So, assertion b is wrong while c is correct.

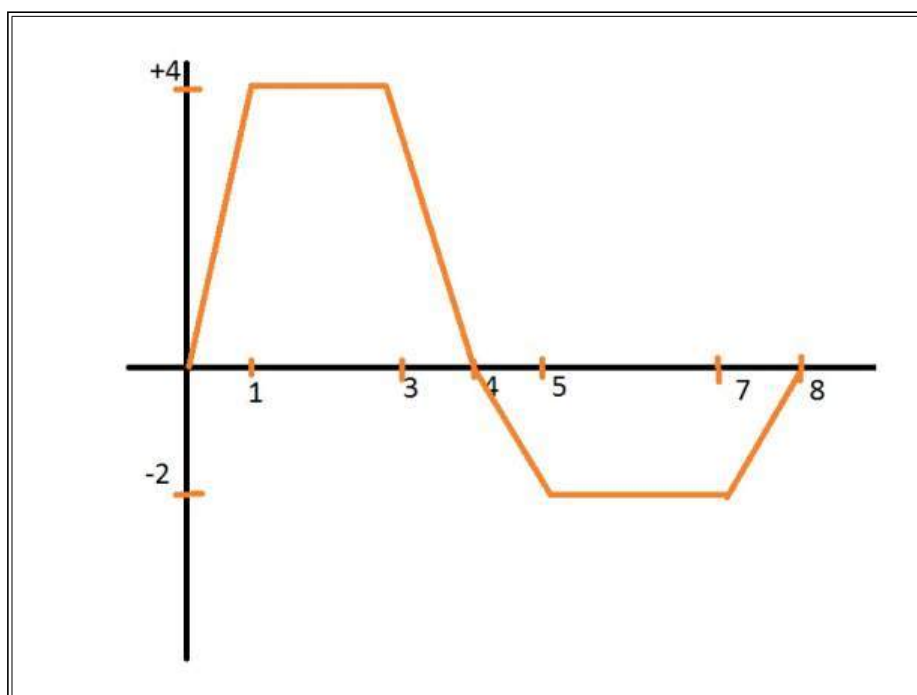


d) As the Graph A has a constant slope, so the acceleration of body A is zero. Whereas Graph B is constantly turning, so the slope can be assumed to be non-zero throughout. According to some revelations, however it is noted that the figure is not clear enough to show whether Graph B is straight after  $t_B$  or bending. In case it is assumed to be straight, then after  $t_B$  both trains will have same (zero) acceleration. Also at start both have large (infinite) acceleration, in which case the ratio of the two large ( infinite ) values may be calculated if initial conditions are mentioned and is required.

At our level we would assume this assertion to be wrong, however making a note that the image should have been more clearly presented.

Answer: c) is the correct assertion. }

**Example:** The velocity-time graph of a particle in linear motion is as shown. Both  $v$  and  $t$  are in SI units. The displacement of the particle is



- a) 6 m
- b) 8 m
- c) 16 m
- d) 18 m

{ Hint : For displacement calculations, between 0 - to 4 , area of the positive trapesium =  $\frac{1}{2} \times (2 + 4) \times 4 = 12$

between 4 - to 8, area of negative trapesium =  $\frac{1}{2} \times (2 + 4) \times (-2) = -6$ .

So , the answer is +6

So , a is the correct option. }

### 1.1.2 The Second Equation

Proceeding similar to above, the equation  $a = \frac{dv}{dt}$  implies

i) The **Slope of Velocity-Time Graph** is **Instantaneous Acceleration**.

ii) The **Area under Acceleration-Time Graph** is **Change in Velocity**.

{ The second one requires the manipulation ,  $dv = a dt$  i.e.  $\int dv = \int a dt$  }

A few of the following examples illustrate it.

**Example:** A car starts from rest acquires a velocity  $v$  with uniform acceleration  $2ms^{-2}$  then it comes to stop with uniform retardation  $4ms^{-2}$  . If the total time for which it remains in motion is 3 sec, the total distance travelled is:

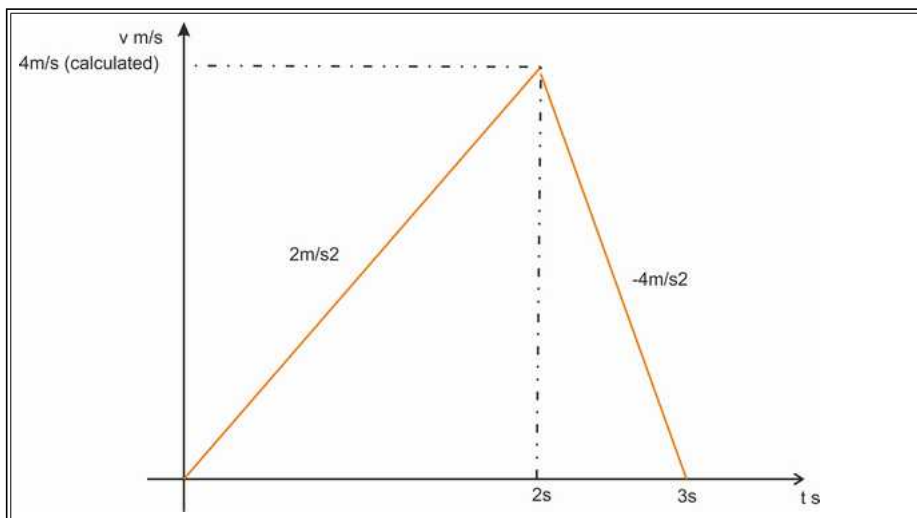
a) 2 m

b) 3 m

c) 4 m

d) 6 m

{Hint: For solving this problem, we draw the graph of the problem,



According to graph, let the time when it reaches maximum velocity be  $T$ , and the maximum velocity be  $V$ .

$$\Rightarrow V = 2XT \text{ and also } V = 4X(3-T)$$

Equating the equations,

$$2T = 12 - 4T = V$$

$$\Rightarrow 6T = 12$$

$$\Rightarrow T = 2$$

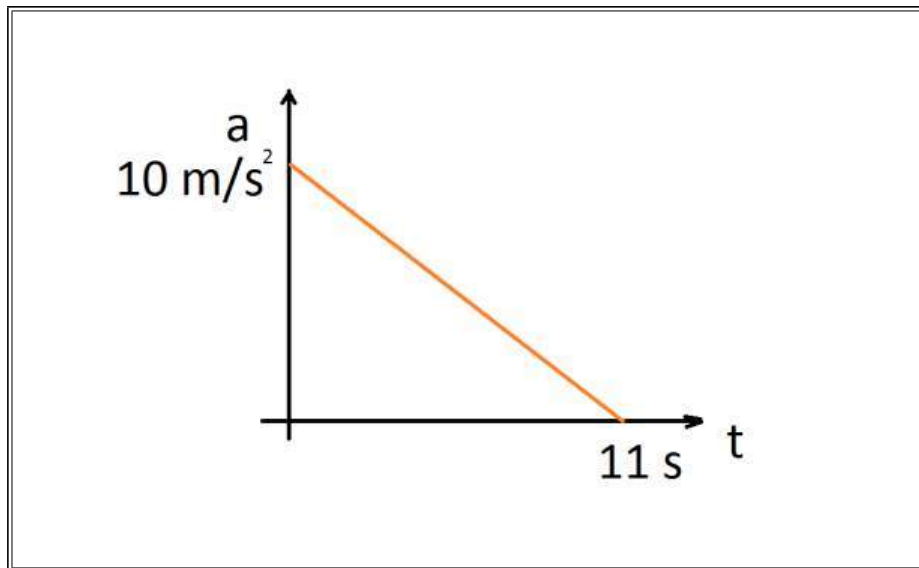
$$\Rightarrow V = 2T = 4$$

Calculating the area under the graph using the calculated parameters, Area =  $\frac{1}{2} \times 4 \times 3 = 6m$

So , area under the graph is  $6m = \text{displacement}$  . Also, as all the area is on the positive side, so distance =  $6m$ .

}

**Example:** A particle starts from rest. Its acceleration (a) vs time (t) is as shown in the Figure. The maximum speed of the particle will be



- a) 110 m/s
- b) 55 m/s
- c) 550 m/s
- d) 660 m/s

{ Hint : Writing the equation of the graph , we get  $\frac{a}{10} + \frac{t}{11} = 1$

$$\Rightarrow a = \frac{10}{11}(11 - t)$$

Integrating, ( we will assume initial velocity to be zero as the body starts from rest. )

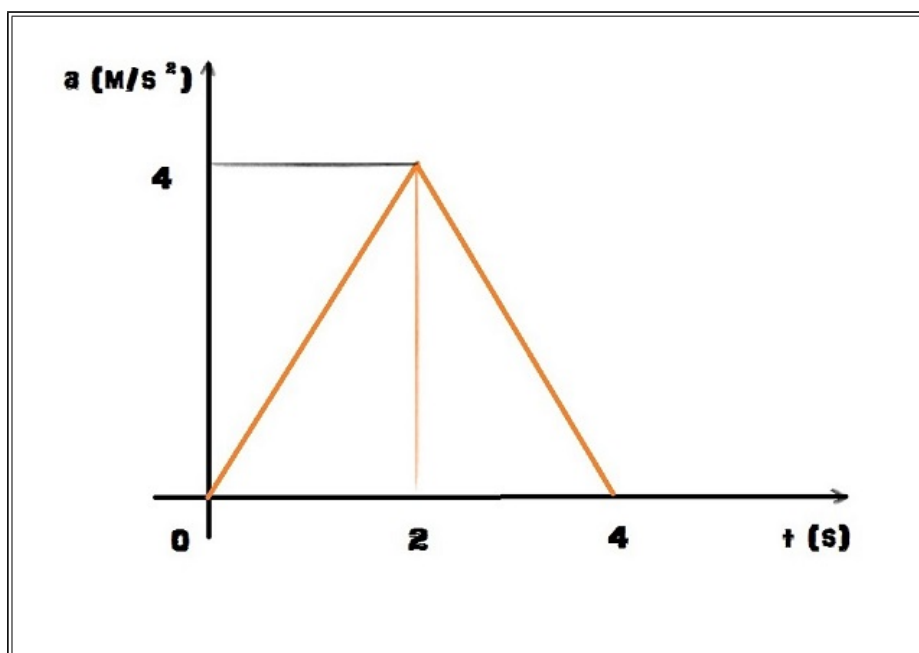
$$v = \frac{10}{11}(11t - \frac{1}{2}t^2)$$

Substituting  $t = 11s$

$$v_{11s} = 55m/s$$

Answer: b) is the correct answer }

**Example:** Acceleration-time graph of a particle moving in a straight line is shown in Figure. The velocity of particle at time  $t = 0$  is 2 m/s. Velocity at the end of fourth second is

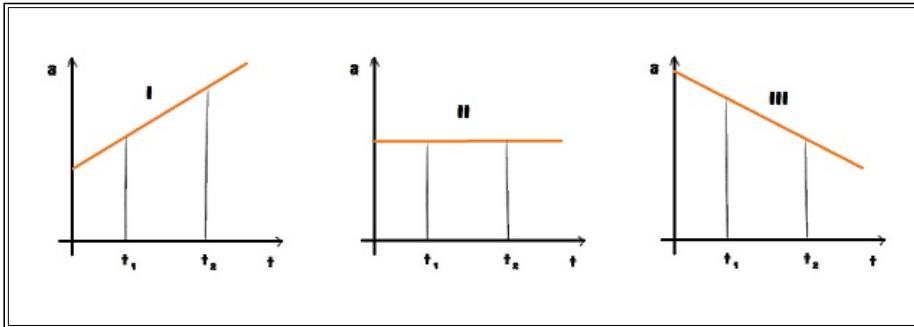


- a) 8 m/s
- b) 10 m/s
- c) 12 m/s
- d) 14 m/s

{ Hint: Area under the acceleration-time graph is change in velocity. Area of the triangle is Half ( into ) base ( into ) altitude = 8m/s. Adding the initial value of 2m/s , we get  $2 + 8 = 10\text{m/s}$ .

Answer: b) is the correct answer }

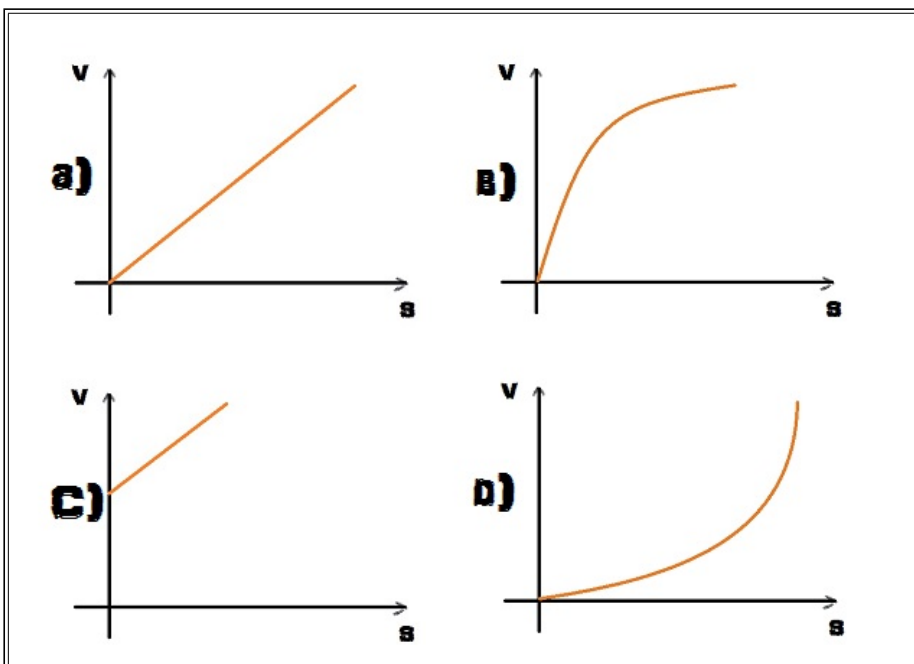
**Example:** Each of the three graphs represents acceleration vs time for an object that already has a positive velocity at time  $t_1$  . Which graph/graphs show an object whose speed is increasing for the entire time interval between  $t_1$  and  $t_2$  ?



- a) Graph I only
- b) Graphs I and II
- c) Graphs I and III
- d) Graphs I, II and III

{Hint: Area under the acceleration time graphs give change in velocity. In all the three figures, the area under the graphs are +ve, hence velocity is increasing in all cases. As the initial velocity is +ve , in all three cases the velocity remains throughout positive. So , the speed is also increasing in all cases.}

**Example:** A body starts from rest and moves along a straight line with constant acceleration. The variation of speed  $v$  with distance  $s$  is given by the graph



{ Hint: The problem given has  $v_o = 0$

Now acceleration = constant ( lets say  $k$  ) =  $\frac{dv}{dt}$

$$\Rightarrow k = v \frac{dv}{ds}$$

$$\Rightarrow \int_0^v v dv = \int_0^s k ds$$

$$\Rightarrow v = \sqrt{2ks}$$

The graph is proportional to square root function

Hence , b (as it is the only graph with such a property )

Answer : b) is the correct graph }

### 1.1.3 The Acceleration-Position Graph Variate

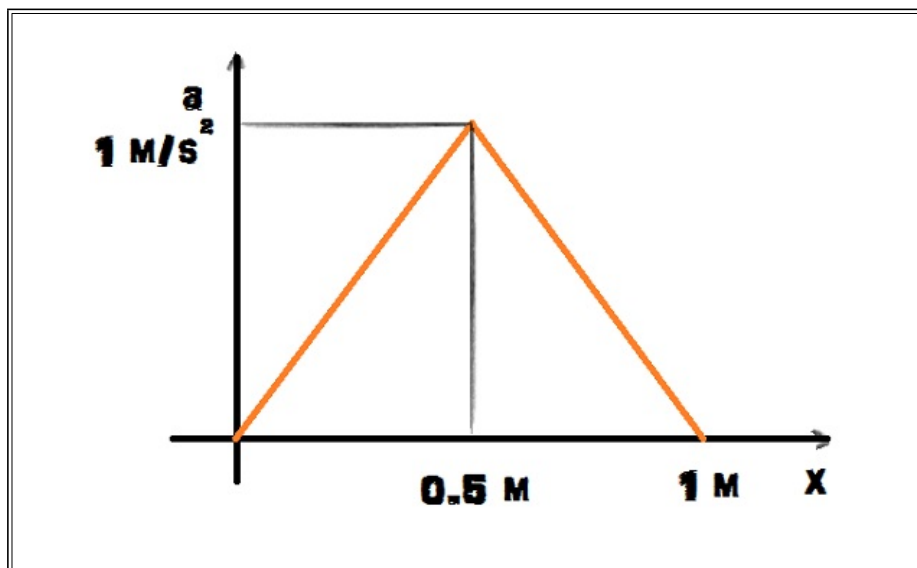
This kind of graph requires the manipulation of the Equation  $a = \frac{dv}{dt}$  as follows

$$a = \frac{dv}{dx} \cdot \frac{dx}{dt}$$

$$\Rightarrow a = \frac{dv}{dx} \cdot v$$

$\Rightarrow a dx = v dv$  and integration can be performed to further solve it.

**Example:** A body, initially at rest, starts moving along x-axis in such a way that its acceleration vs displacement plot is as shown in the Figure. The maximum velocity of the particle is



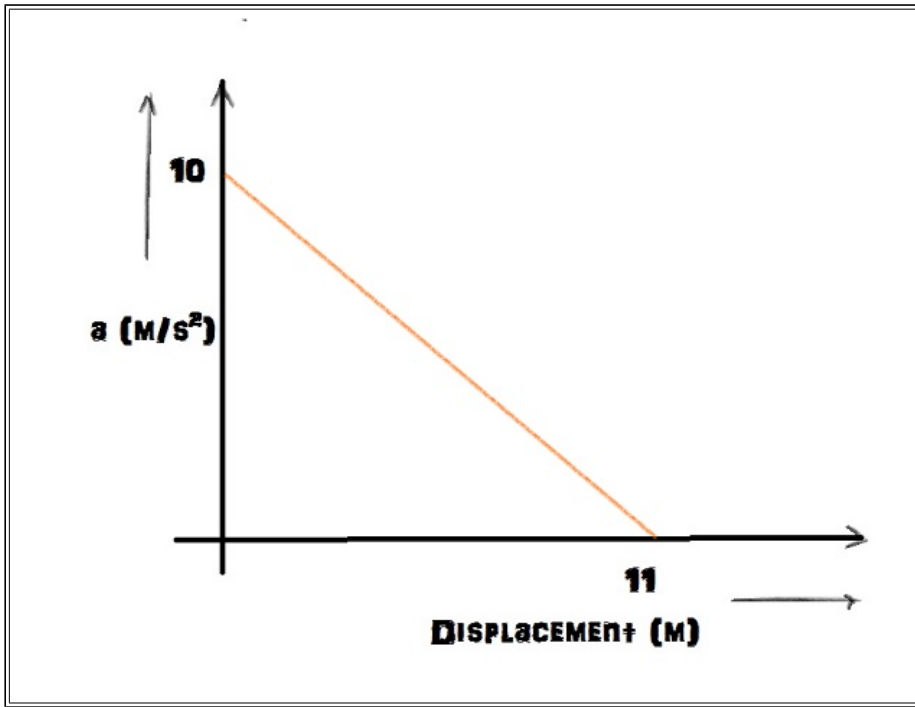
- a) 1 m/s
- b) 6 m/s
- c) 2 m/s
- d) None of these

{ Hint : The area under the a-x graph gives change in  $\frac{v^2}{2}$  as can be evaluated by Integration method.  
The area of triangle is  $(0.5) \times (1) \times (1) = 0.5$

$\Rightarrow v = 1 \text{ m/s}$  . Initial position is given to be 0 in the graph. Hence, we take only the positive sign.

Answer : a) is the correct answer. }

**Example :** A particle initially at rest, it is subjected to a non-uniform acceleration a, as shown in the gure. The maximum speed attained by the particle is



- a) 605 m/s
- b) 110 m/s
- c) 55 m/s
- d) 110 m/s

{ Hint: The area under the a-t graph is half the change in  $v^2$ . Area under the graph is 55. So  $v_{max} = \sqrt{110} \text{ m/s}$ . Hence none of the answers is correct }

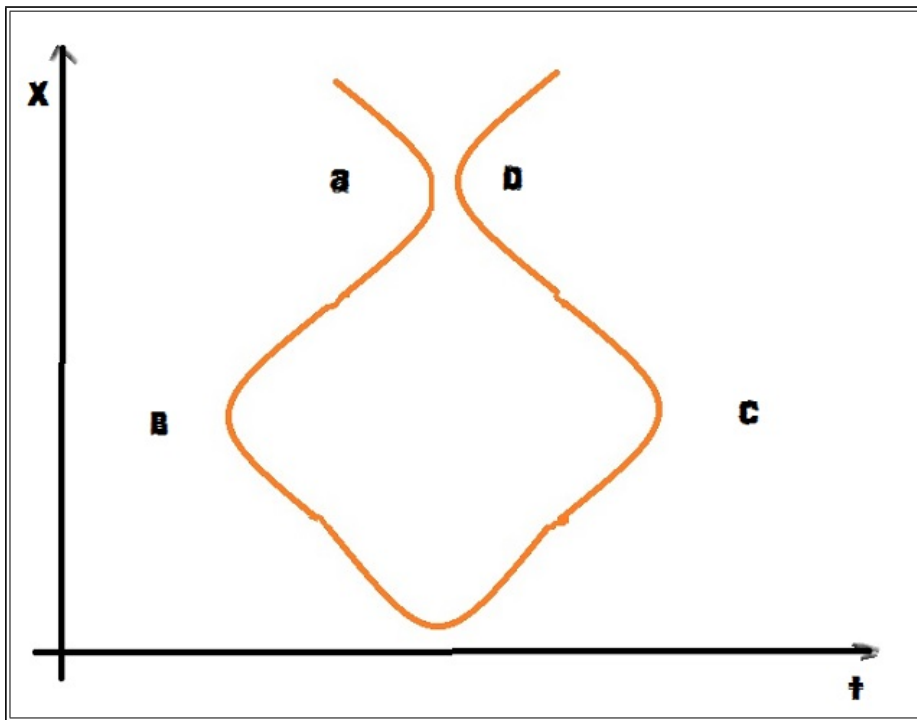
## 1.2 Other Types of Graphs

### 1.2.1 Sign of Acceleration from Position-Time Graph

The sign of Acceleration can be determined from the Position-Time Graph. The methodology involves looking at the Concavity of the Graph

- i) If the graph is Concave-Up, the Acceleration is Positive.
- ii) If the graph is Concave-Down, the Acceleration is Negative.
- iii) If the graph is a straight line, the Acceleration is ZERO. { Irrespective of any other factor , such as the slope or direction of line }

**Example:** The graph given below is a plot of distance vs time. For which labelled region is the “Velocity Positive and the Acceleration Negative”



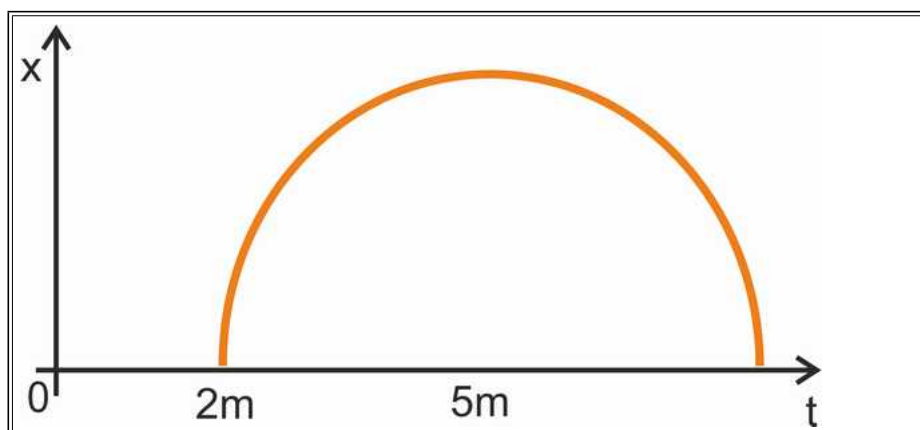
- a) a
- b) b
- c) c
- d) d

{ Hint: By the above mentioned propositions, d) is the required section with positive velocity and negative acceleration. }

### 1.2.2 The Average-Velocity / Instantaneous Velocity , Equal Case

We know , that ( in a x-t graph) the slope of the Secant is the Average Velocity , whereas the slope of Tangent is the Instantaneous Velocity. The point where these two lines coincide, is the point where Average Velocity is equal to Instantaneous Velocity.

**Example:** Position-time graph is shown which is a semicircle from  $t = 2$  to  $t = 8$  s. Find time  $t$  at which the instantaneous velocity is equal to average velocity over first  $t$  seconds,



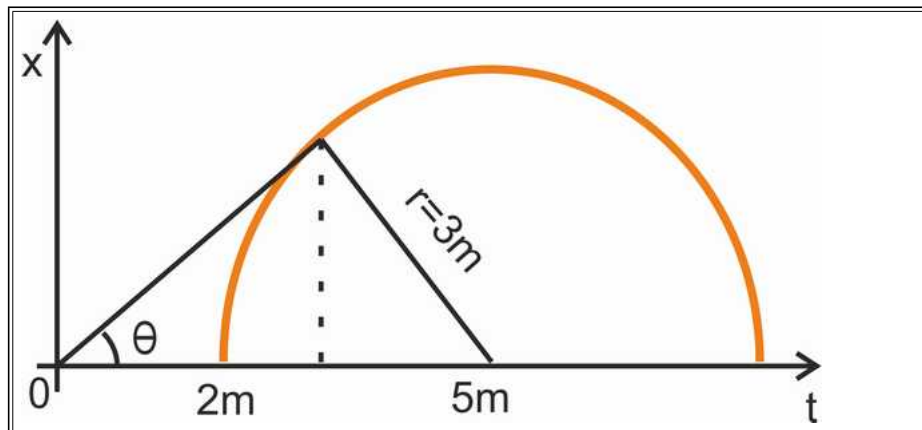
- a) 4.8 s
- b) 3.2 s
- c) 2.4 s
- d) 5 s

{ Hint: The tangent from 0 to the circle is drawn. It's normal passes through the center of the circle. Time at this instant needs to be calculated. }

If  $H=5$  ,  $R = 3$  , Length of tangent = 4. (By Pythagoras.)

Angle which the tangent makes with the  $t$  axis is  $\theta = \sin^{-1}(3/5)$

So, the projection of tangent on  $t$  axis ( i.e. the required time ) =  $4 \cos \theta = 4 \times \frac{4}{5} = 3.2$



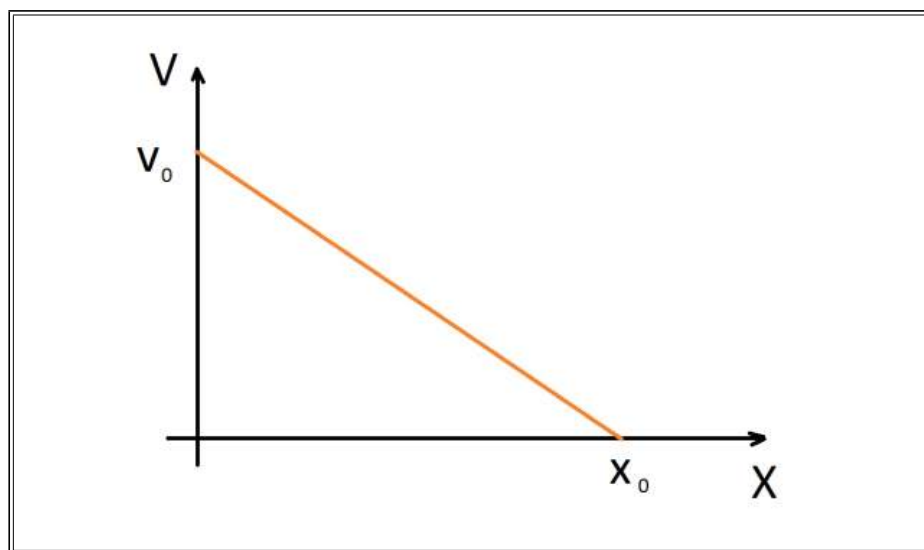
}

### 1.2.3 The Velocity-Displacement Case

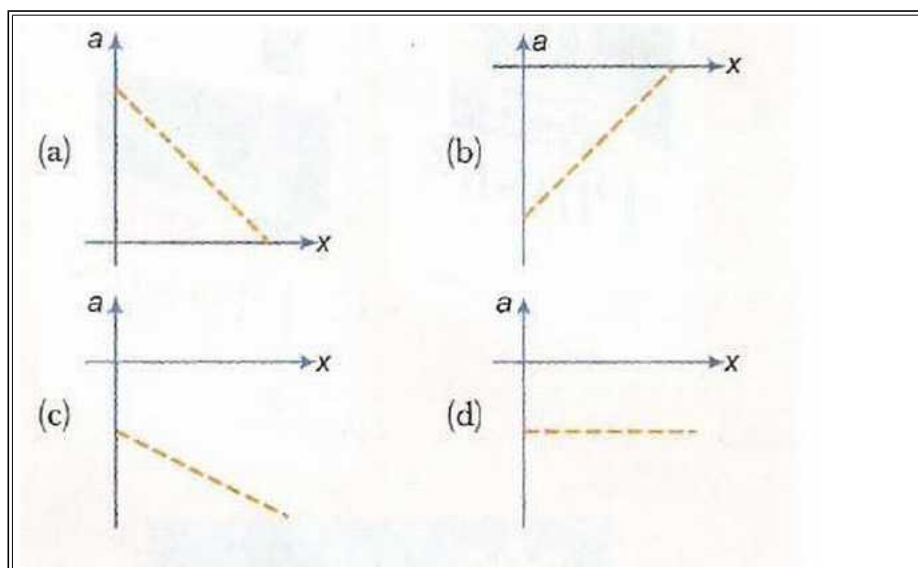
This can be handled in a similar way as Acceleration-Displacement case by integrating the respective equation. Here the problem is of  $v = f(x)$  type, which can be integrated by writing  $\frac{dx}{dt} = f(x)$

i.e.  $dx = f(x)dt$

**Example:** The velocity-displacement graph of a particle moving along a straight line is shown here.



The most suitable acceleration-displacement graph will be



{Hint: Using Co-Ordinate Geometry Result studied in +1 Mathematics, we get the equation of the graph

$$\frac{v}{v_o} + \frac{x}{x_o} = 1$$

We are supposed to find the a-x graph from this.

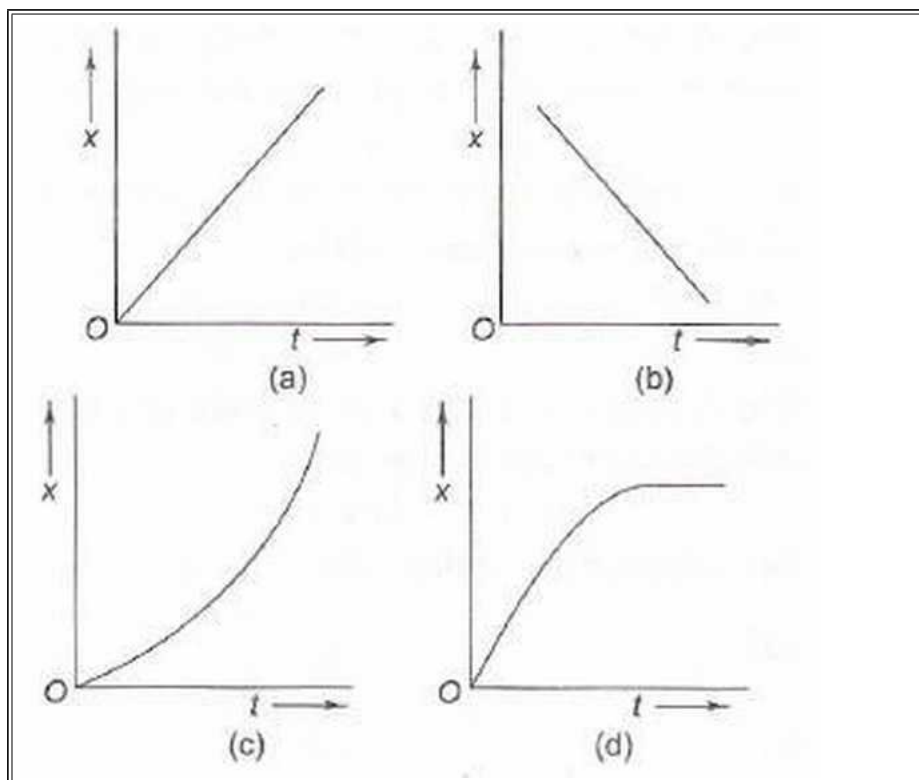
So, we rewrite this equation as  $v = v_o(1 - \frac{x}{x_o})$

Differentiating, we get  $a = -\frac{v_o}{x_o} = \text{constant}$

Hence d) is the requisite graph.

Answer: d) is the correct answer. }

**Example.** The velocity (v) of a body moving along the positive x-direction varies with displacement (x) from the origin as  $v = kx$ , where k is a constant. Which of the graphs shown in Fig. correctly represents the displacement-time (x - t) graph of the motion?



{Hint: The variable p removes the dependence of v on x and gives emphasis only on the first condition that body is moving along positive x-direction. In Graphs a) ,c) body is moving along

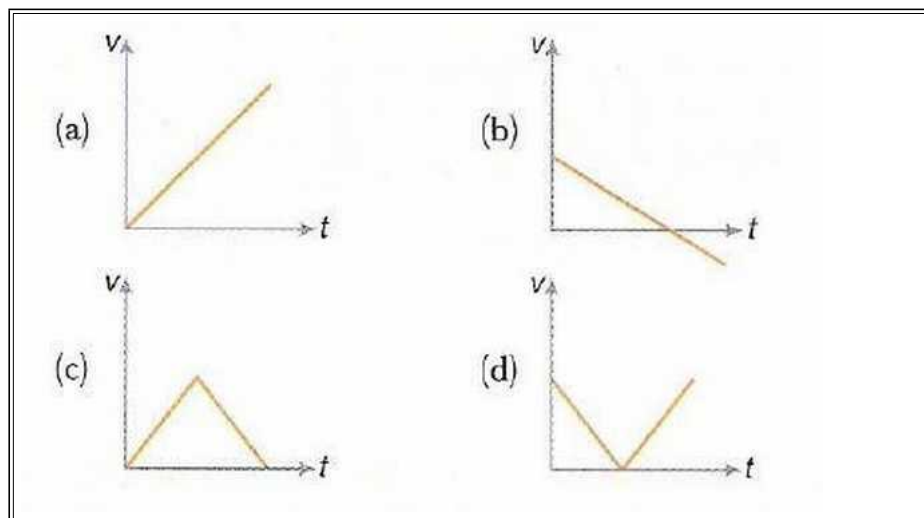
positive x-direction, However, a) is a specific case when  $p$  is proportional to  $x$  and not the general case. Only c) covers the general case of all possible  $p$  and still moving in positive x-direction. Hence c) is the correct answer

Answer: c) is the required answer. }

### 1.2.4 Motion Under Free Fall due to Gravity

In such examples, the governing equations rule and the coordinate system needs to be properly chosen.

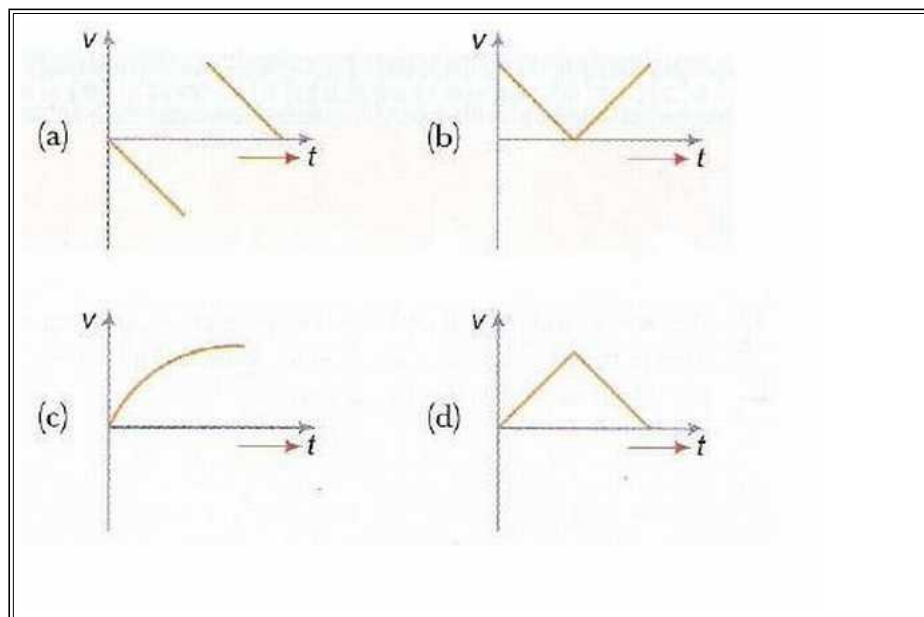
**Example:** Which of the following graphs correctly represents velocity-time relationship for a particle released from rest to fall freely under gravity?



{ Hint: If we take the downward axis as positive,  $v$  will keep on increasing till the object hits something.

So, a) is the correct answer.}

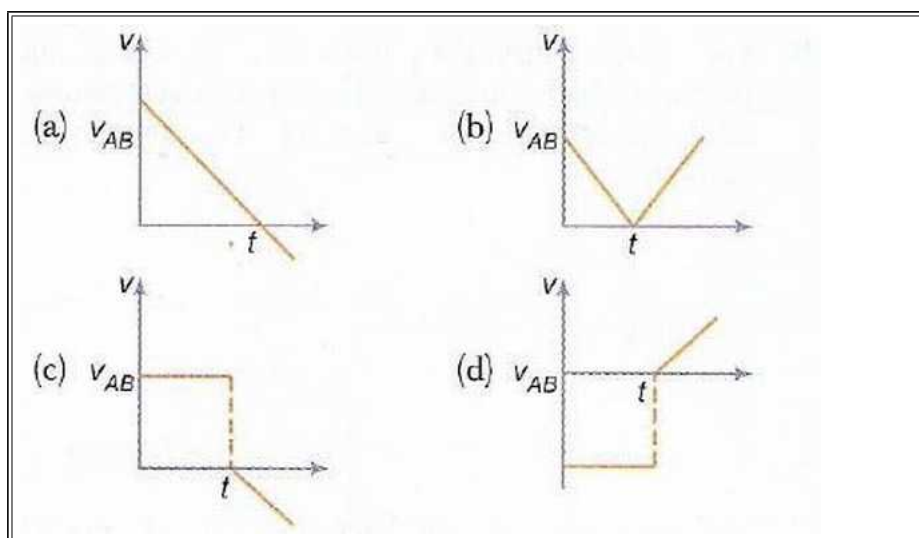
**Example:** The velocity-time graph of a body falling from rest under gravity and rebounding from a solid surface is represented by which of the following graphs?



{ Hint: At rebound, the velocity would become negative of itself if the collision is perfectly elastic. Only in graph (a) such a thing is happening.

a) is the correct answer.}

**Example:** A body A is thrown vertically upwards with such a velocity that it reaches a maximum height of  $h$ . Simultaneously another body B is dropped from height  $h$ . It strikes the ground and doesn't rebound. The velocity of A relative to B vs time graph is best represented by (upward direction is positive.)

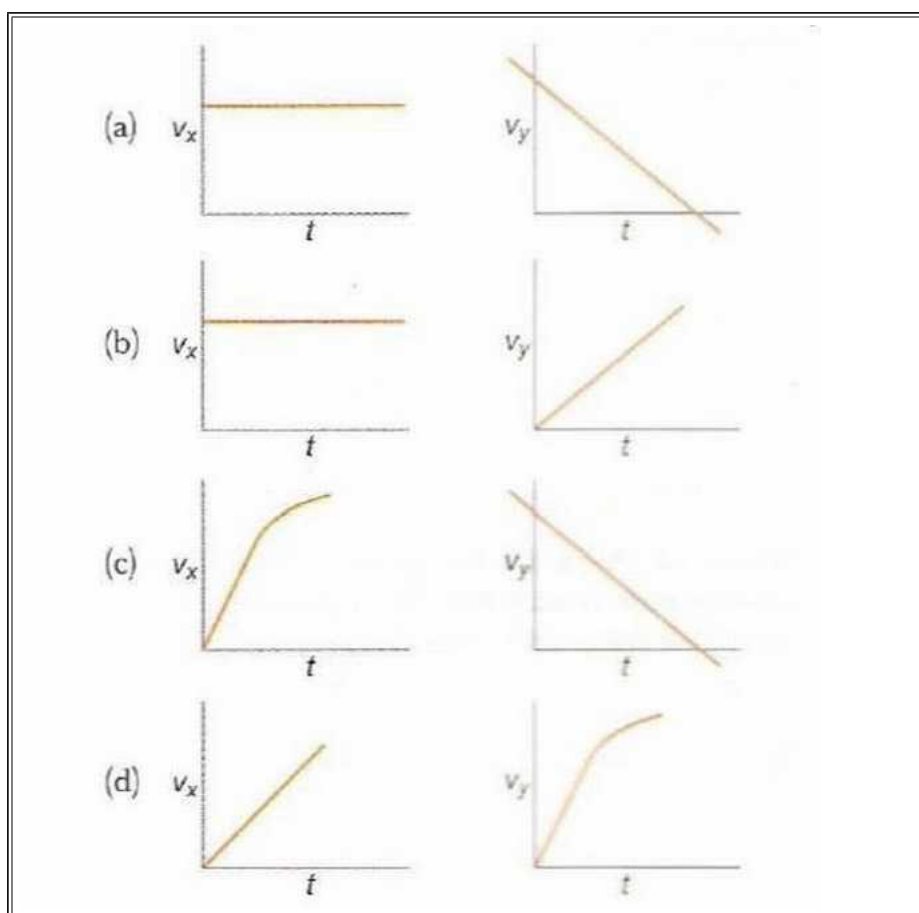


{ Hint: Before the strike, body A has velocity  $u - gt$  whereas body B has velocity  $-gt$ . So, a constant difference of  $u$  remains. Before strike,  $u - gt - (-gt) = u$ . However, after that it is  $u - gt$ , the time being  $t_0 = u/g$  of strike. So, the negative slope line starts from the x-axis and a discontinuity comes into picture.

So, C) is the required graph. }

### 1.2.5 Projectile Motion

**Example:** A shell is fired from a gun at an angle to the horizontal. Graphs are drawn for its horizontal component of velocity  $v_x$  and its vertical component of velocity  $v_y$ .

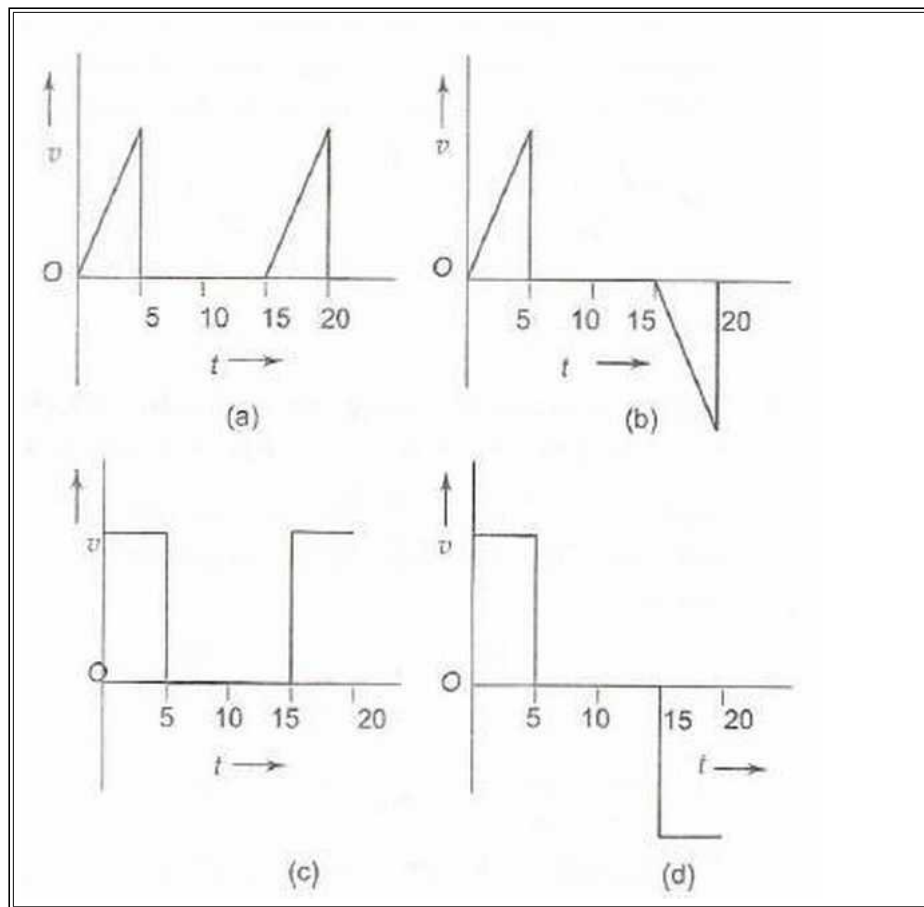
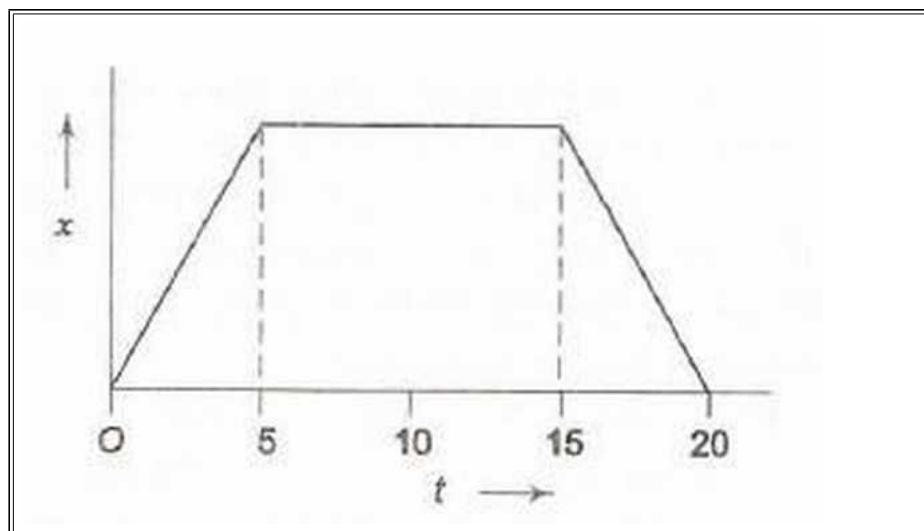


{ Hint:  $v_x$  would remain constant as  $u \cos \theta$  and  $v_y$  would linearly decrease and go negative after maximum height is attained (if the fire angle is +ve).

So, a) is the requisite graph. }

### 1.2.6 Miscellaneous

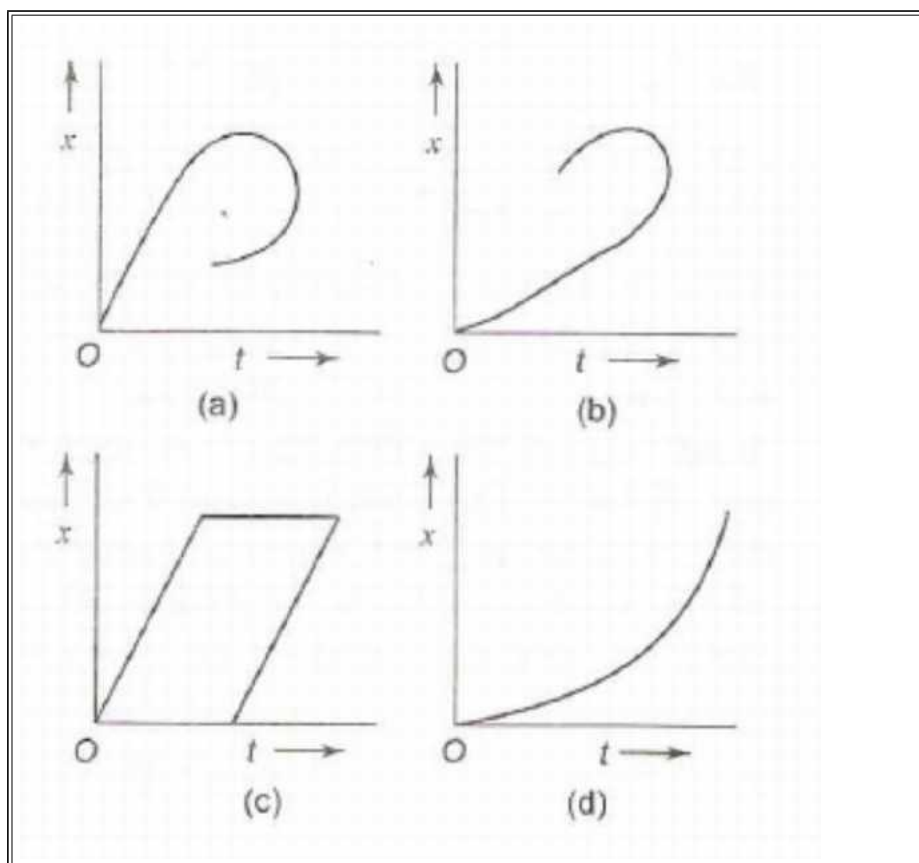
**Example.** Figure shows the displacement-time ( $x-t$ ) graph of body moving in a straight line. Which one of the graphs shown in Fig. represents the velocity-time ( $v-t$ ) graph of the motion of the body.



{ Hint: The graph has initially (0-5) a +ve slope, then zero slope and then (15-20) an equal -ve slope

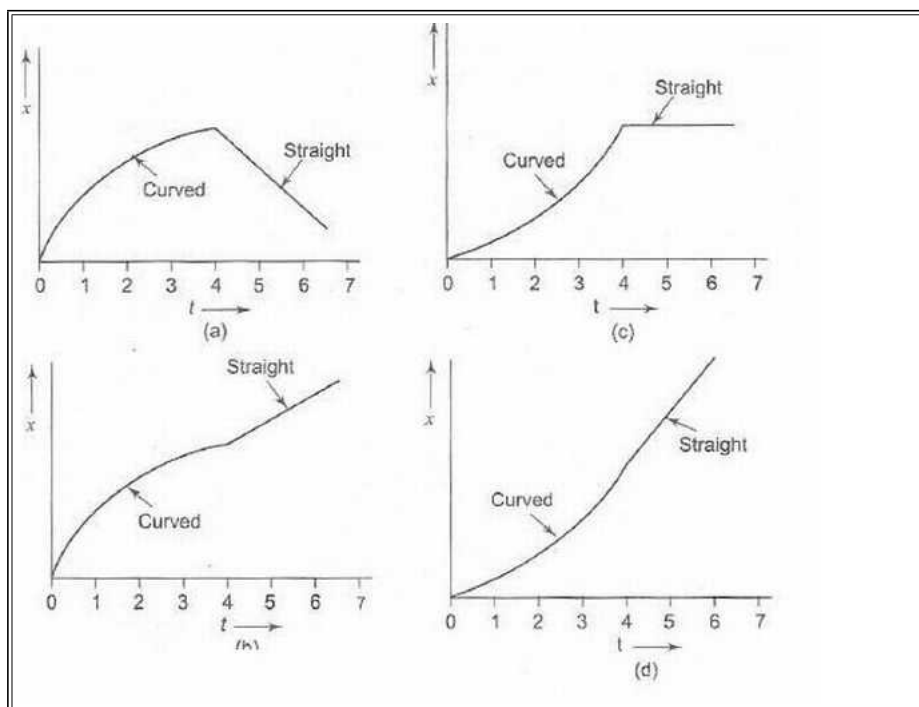
So, d) is the correct answer. }

**Example.** Which of the displacement-time ( $x-t$ ) graphs shown in Fig. can possibly represent one dimensional motion of a particle?



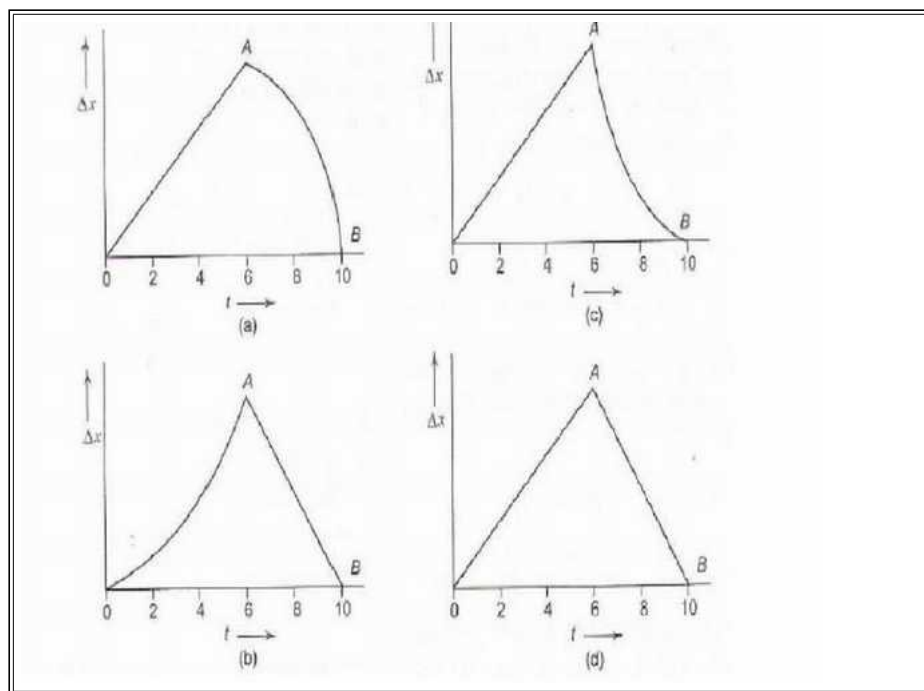
{ Hint: The object cannot be at two positions on a single time instant,  
So, d) is the correct option. }

**Example.** A car starts from rest, accelerates uniformly for 4 seconds and then moves with uniform velocity.  
Which of the ( $x-t$ ) graphs shown in Fig. represents the motion of the car upto  $t = 7$  s?



{ Hint: As the car is accelerating upto 4 seconds, the graph should be concave up. Further ahead ,  
it moves with constant velocity , so a positive slope ahead of 4 seconds.  
d) is the correct answer. }

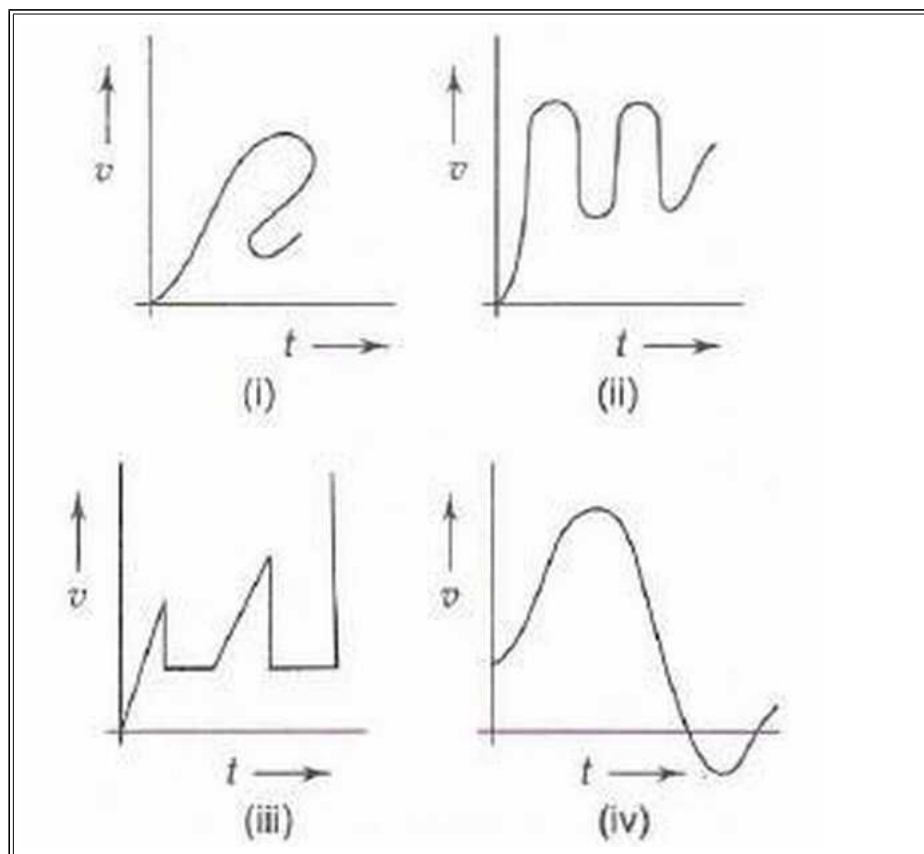
**Example.** Two stones are thrown up simultaneously with initial speeds of  $u_1$  and  $u_2$ , ( $u_2 > u_1$ ). They hit the ground after 6 s and 10 s respectively. Which graph in Fig. correctly represents the time variation of  $\Delta x = x_2 - x_1$ , the relative position of the second stone with respect to the first upto  $t = 10$  s? Assume that the stones do not rebound after hitting the ground.



{ Hint : While both are in air, the difference of their velocity vectors would be a constant vector , so  $\Delta x = c_1 t$  . Also, after the first stone hits ground,  $\Delta x = c_2 t - x_o$  as the velocity component in x direction of second stone is constant. There is no discontinuity. So, both before and after one hits the ground , it would be a straight line.

Hence, d) is the correct answer. }

**Example.** Figure shows the velocity-time ( $v - t$ ) graphs for one dimensional motion. But only some of these can be realized in practice. These are

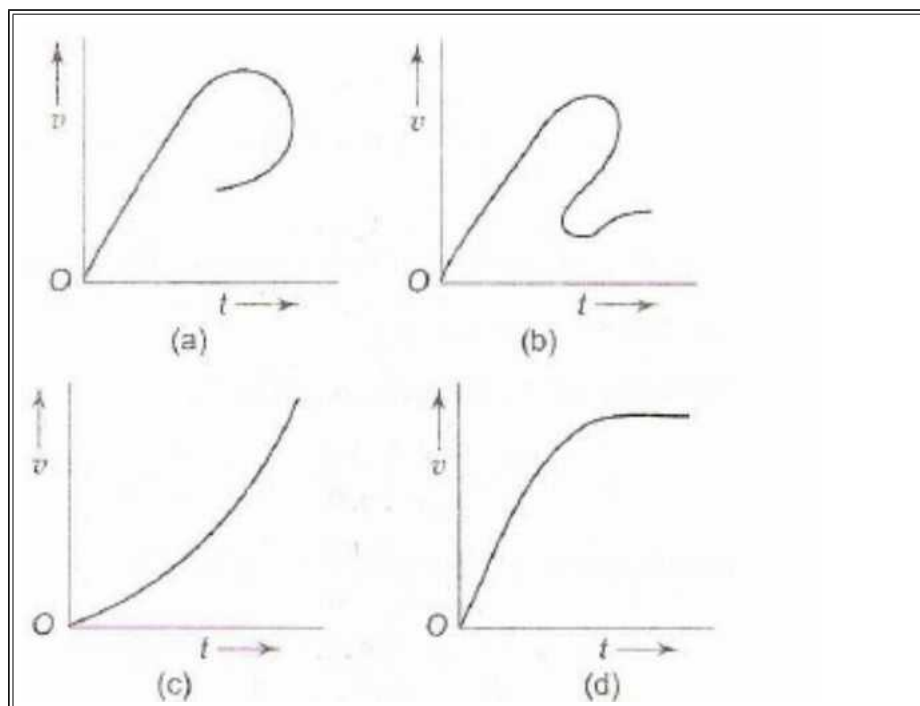


- a) (i), (ii) and (iv) only
- b) (i), (ii) and (iii) only
- c) (ii) and (iv) only
- d) all

{ Hint: At one particular instant, the object cannot have two different velocities. }

Hence, c) is the correct answer. }

**Example.** Which of the velocity-time (v-t) graphs shown in Fig. can possibly represent one-dimensional motion of a particle?

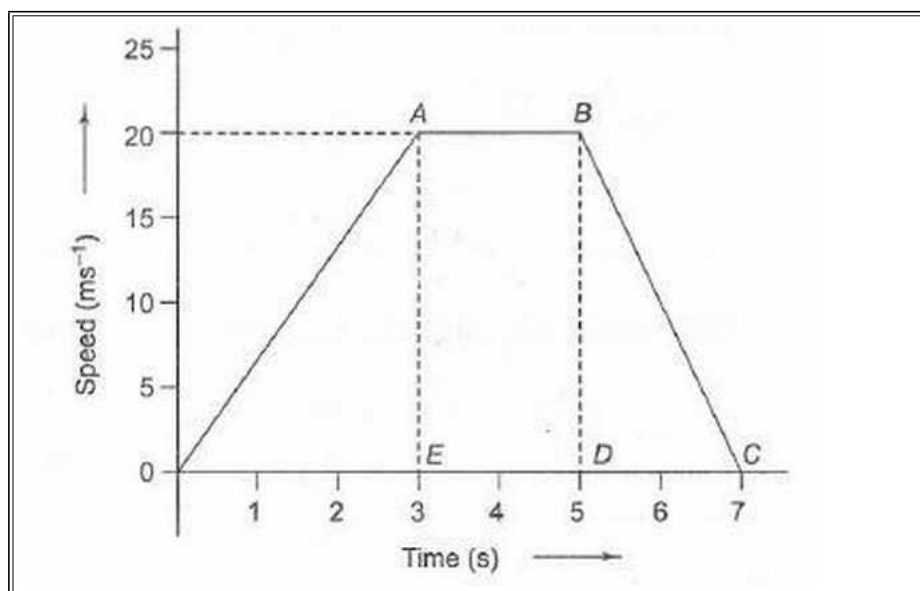


{Hint: In c) , the particle is constantly increasing in one dimension, In d) , it stops near the end. c),d) are the correct answers. }

## 1.3 Question Types

### 1.3.1 Passage Type

**Example:** The speed-time graph of the motion of a body is shown in Fig.



1. The accelerations of the body during the last 2 second is

- a)  $\frac{20}{3}ms^{-2}$
- b)  $-\frac{20}{7}ms^{-2}$
- c)  $-10ms^{-2}$
- d) Zero

2. The ratio of distance travelled by the body during the last 2 seconds to the total distance travelled by it is

- a) 1/9
- b) 2/9
- c) 3/9
- d) 4/9

3. The average speed of the car during the whole journey is

- a) 10 m/s
- b) 20 m/s
- c)  $90/7$  m/s
- d)  $40/7$  m/s

{Hint: 1. As no contradictory statements are present, we would take velocity as the value of speed only. Actually velocity is required for calculation of acceleration but here , speed doesn't contradict anything so we'll use it's value for velocity.

-10 from slope, c)

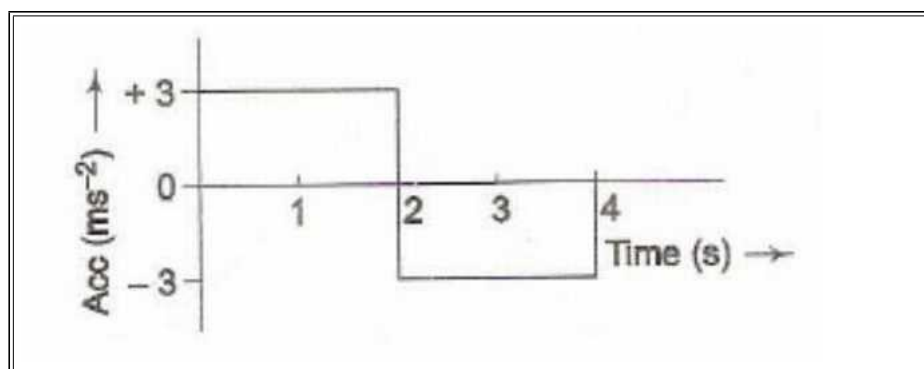
2. Speed time graph's area is distance. From area calculation, in last 2 seconds, the area is 20 and total area is 90.

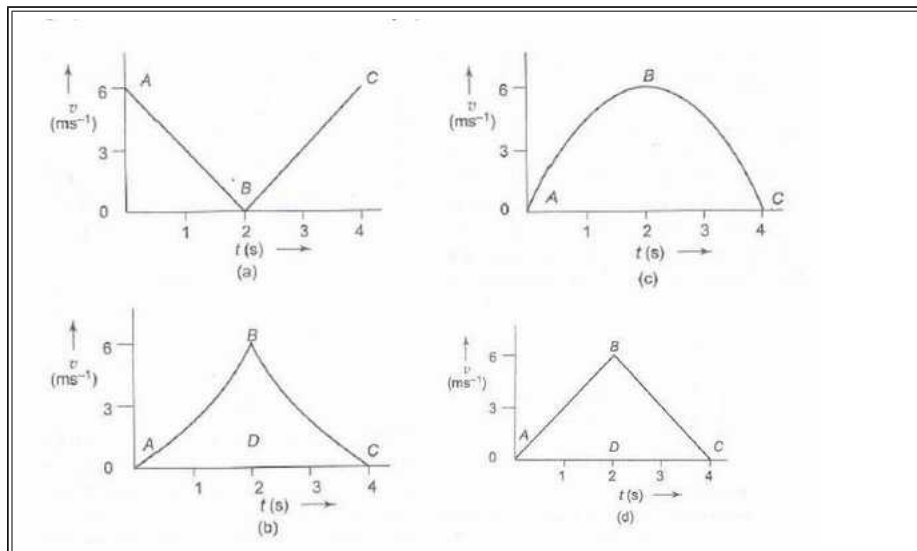
So, 2/9 is the required ration b)

3. Total distance from area is 90, and total time is 7.

So,  $90/7$ . c)}

**Example:** A body starts from rest at time  $t = 0$  and undergoes an acceleration as shown in Fig. Which of the graphs shown in Fig. represents the velocity-time ( $v-t$ ) graph of the motion of the body from  $t = 0$  s to  $t = 4$  s?





{Hint: d) is the v-t graph by calculating area function.}

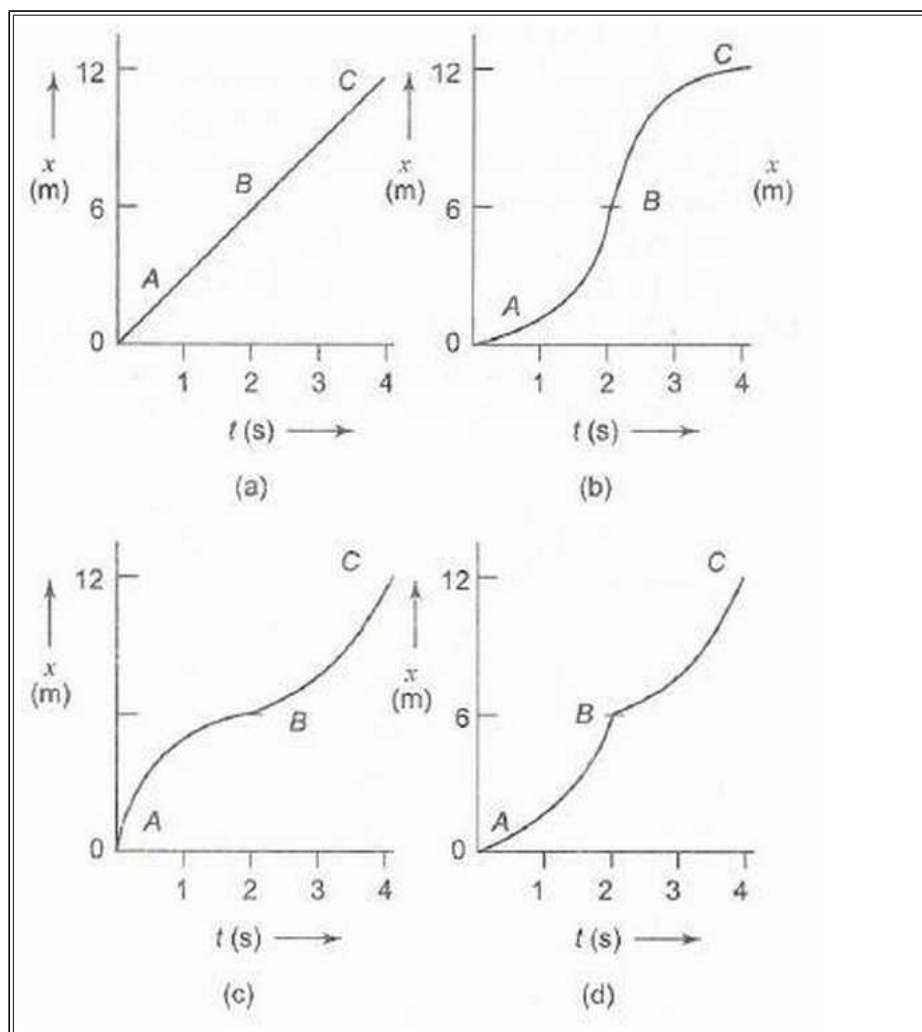
1. In Question above, what is the velocity of the body at time  $t = 2.5$  s?

- a) 2.5 m/s
- b) 3.5 m/s
- c) 4.5 m/s
- d) 5.5 m/s

2. In above question, how much distance does the body cover from  $t = 0$  s to  $t = 4$  s?

- a) 6 m
- b) 9 m
- c) 12 m
- d) 15 m

3 In above question, which of the graphs shown in Fig. represents the displacement-time (x-t) graph of the motion of the body from  $t = 0$  s to  $t = 4$  s?



{ Hint: 1. The area under the accln-time graph gives change in velocity. Till 2.5s,  $6 - 1.5 = 4.5$  is the required area.

So, c)

2. Area under the v-t graph d) is 12 till 4 seconds.

So, c)

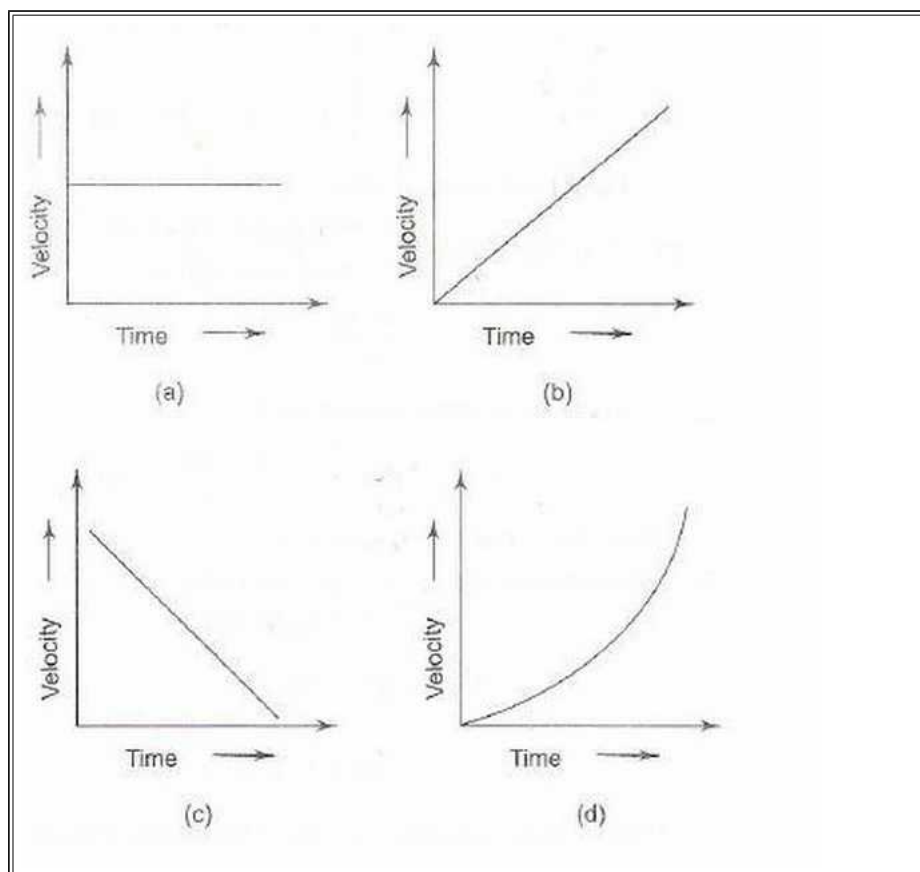
3. Till 2 seconds , the graph should be concave-up while from 2 to 4 it should be concave-down.

So, b)

}

### 1.3.2 Matching

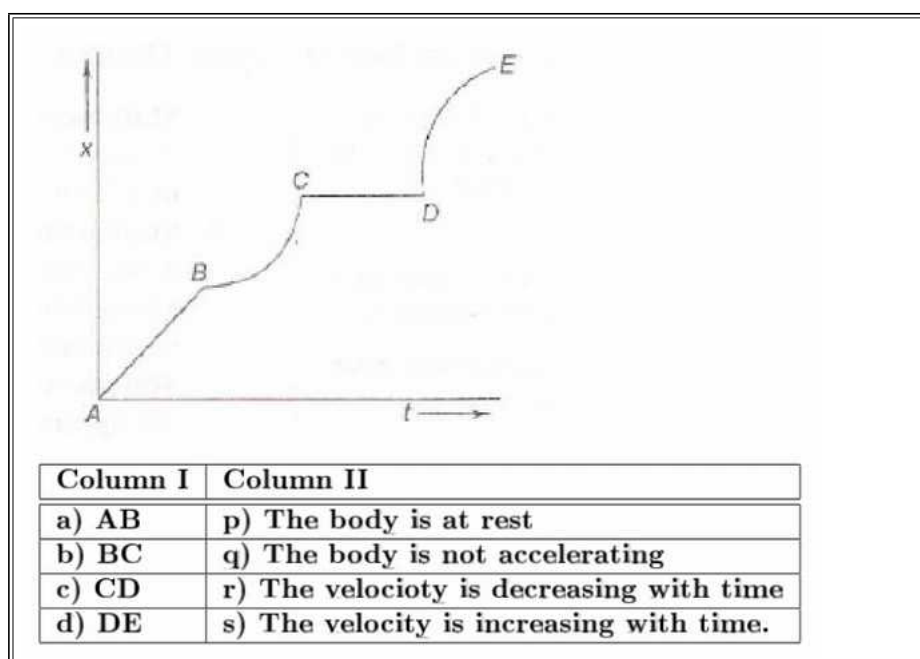
- Match the graphs (a), (b), (c) and (d) shown in Fig. with the types of motions (p), (q), (r) and (s) that they represent



- p) Motion with non-uniform acceleration  
 q) Motion of a body covering equal distances in equal intervals of time  
 r) Motion having a constant retardation  
 s) Uniformly accelerated motion.

{ Hint: p  $\rightarrow$  d , q  $\rightarrow$  a, r  $\rightarrow$  c , s  $\rightarrow$  a,b,c }

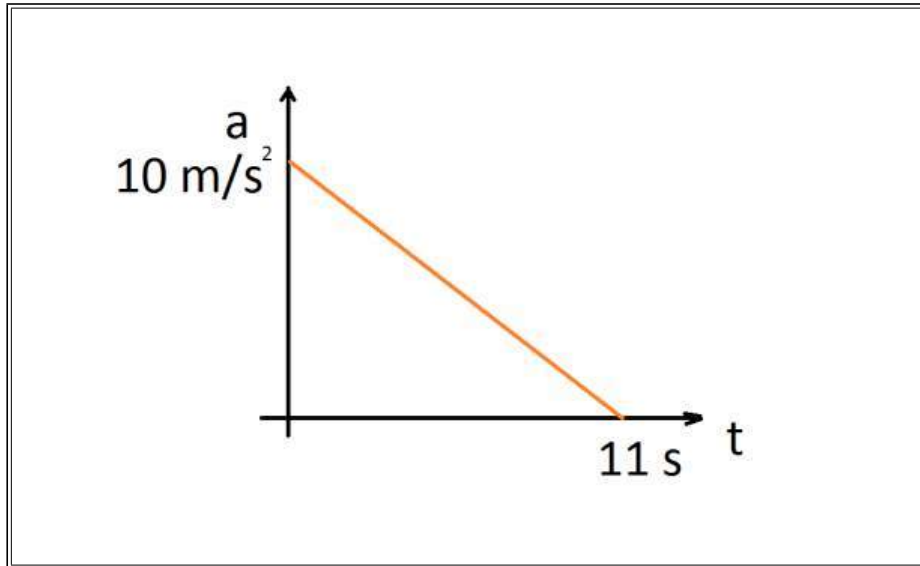
2. Figure shows the displacement-time ( $x-t$ ) graph of the motion of a body.



{ Hint p  $\rightarrow$  c, q  $\rightarrow$  a,c, r  $\rightarrow$  d , s  $\rightarrow$  b }

## 1.4 Previous Year Problems IIT

**Q1:** A particle starts from rest. Its acceleration (  $a$  ) versus time (  $t$  ) is as shown in the gure. The maximum speed of the particle will be



a )  $110 \text{ m/s}$

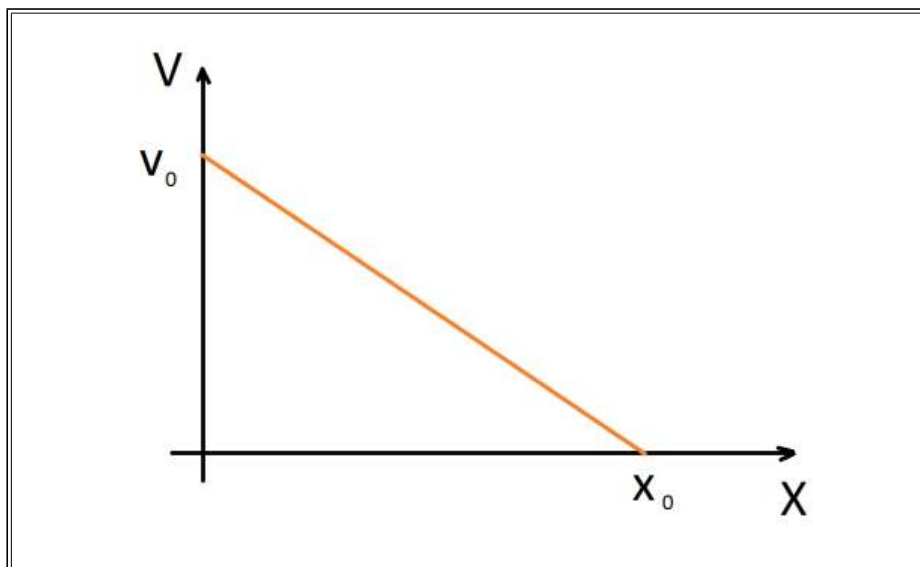
b )  $55 \text{ m/s}$

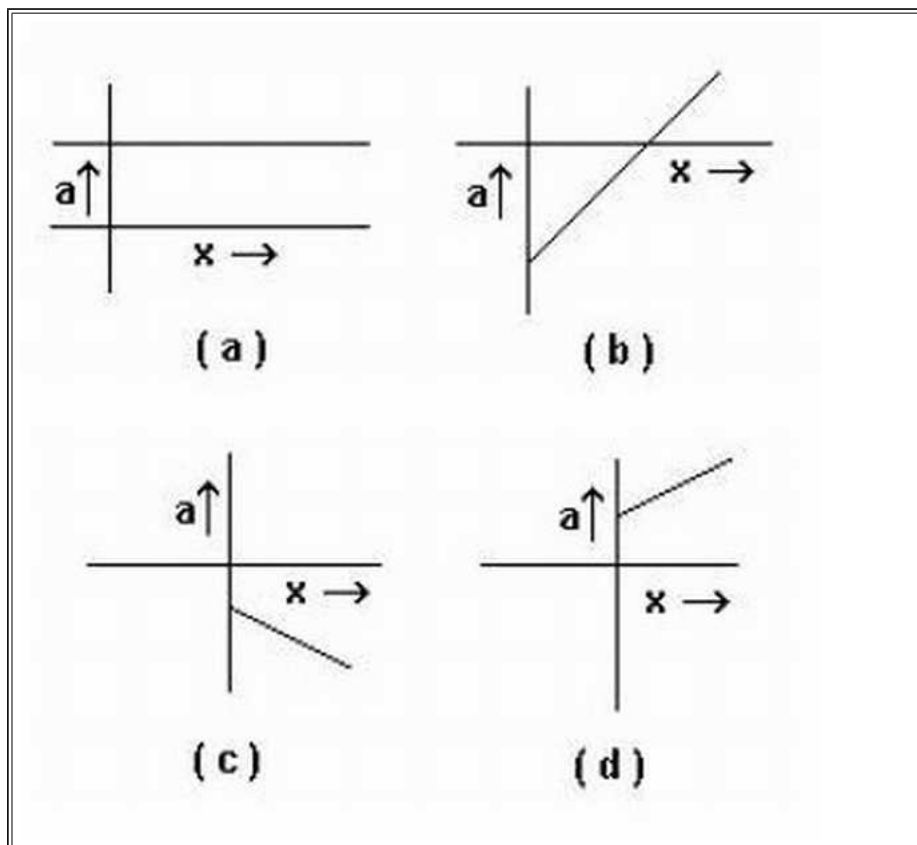
c )  $550 \text{ m/s}$

d )  $660 \text{ m/s}$

{ Hint : See In chapter examples for solution. }

**Q2:** If graph of velocity vs. distance is as shown, which of the following graphs correctly represents the variation of acceleration with displacement ?





{ Hint : The graph of the question is a straight line with the equation,  $\frac{x}{x_o} + \frac{v}{v_o} = 1$

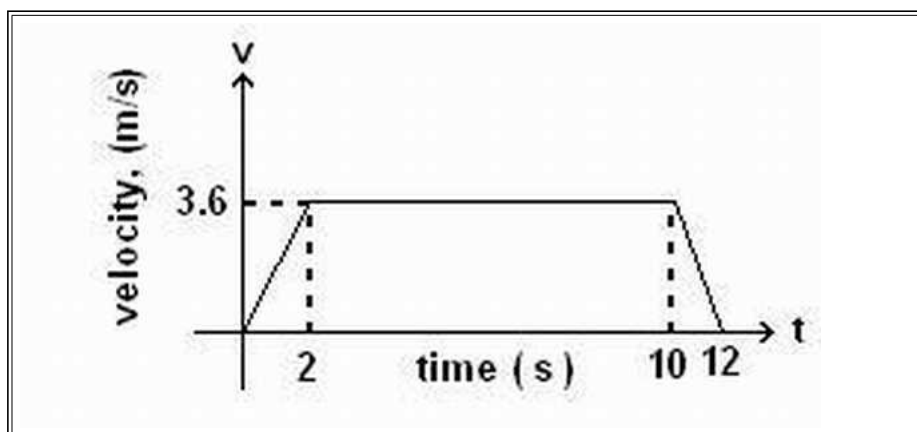
This gives ,  $v = v_o(1 - x/x_o)$

So, differentiating it, we get

$a = -v_o/x_o$  which is a constant. Only in a) it is shown to be a constant.

So, a) }

**Passage** A lift is going up. The variation in the speed of the lift is as given in the graph.



**Q3:** What is the height to which the lift takes the passengers ?

- a ) 3.6 m
- b ) 28.8 m
- c ) 36 m
- d ) cannot be calculated from the above graph

**Q4:** In the above graph, what is the average velocity of the lift?

- a ) 1 m/s
- b ) 2.88 m/s
- c ) 3.24 m/s
- d ) 3 m/s

**Q5:** In the above graph , what is the average acceleration of the lift?

- a )  $1.8m/s^2$
- b )  $-1.8m/s^2$
- c )  $0.3m/s^2$
- d ) zero

{Hint: Q3: Area under the graph is 36. Taking initial position as zero, c) is the best fit answer.

Q4. For av. velocity, total displacement / total time should be calculated. which is  $36/12 = 3m/s$ . d ) is the required answer.

Q5. The accln is 1.8 from 0s to 2s, 0 from 2s to 10s and -1.8 from 10s to 12s. So, area under the a-t graph is zero. So av. accln is zero d)

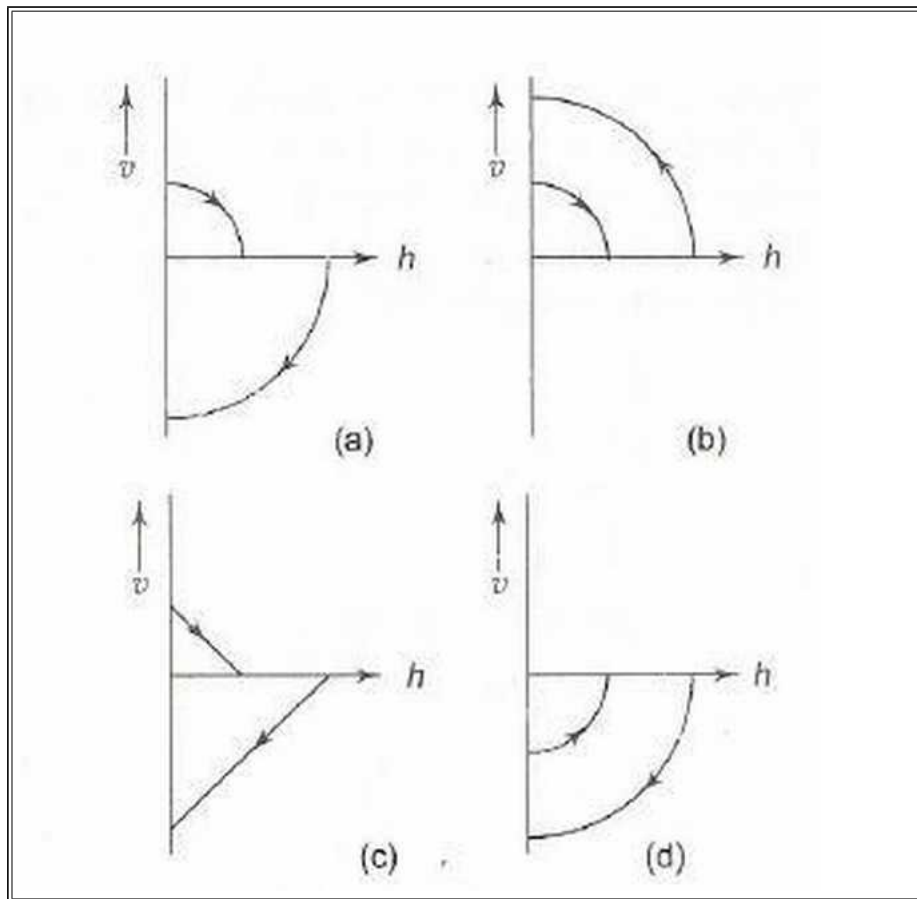
}

**Q6:** Four persons K, L, M and N are initially at the corners of a square of side of length d. If every person starts moving with velocity v such that K is always headed towards L, L towards M, M towards N and N towards K, then the four persons will meet after

- a )  $d/v$  s
- b )  $d^2/v$  s
- c )  $d / 2v$  s
- d )  $d / 2v$  s

{Hint: Not a graphs question, we'll discuss it in kinematics book. Although an easy one. It was mistakenly added to graphs book. Let's use it to signify the fact that even if a diagram is made, it is still not a graph.}

**Example.** A ball is dropped vertically from a height h above the ground. It hits the ground and bounces up vertically to a height h/2. Neglecting subsequent motion and air resistance, its velocity v varies with the height h as (see Fig.) (I.I.T. 2000)



{ Hint : Case 1: Thrown from height  $h$  with zero initial velocity,  $v = -gt$  and  $h = h_o - 1/2gt^2$ , so eliminating  $t$ , we get  $h = h_o - v^2/2g$ . So,  $v = -\sqrt{2g(h_o - h)}$

Case 2: We assume the opposite for solving purpose that the ball is now thrown from a height  $h_o/2$  and replace  $h_o$  only and see the effect. Taking upward velocity as positive, we get  $v = \sqrt{2g(h_o/2 - h)}$

a) is the required answer. }